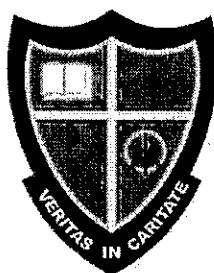


ST BENEDICT'S COLLEGE



SUBJECT	Advanced Programme Mathematics Paper 2	DATE	November 2014
	STATISTICS		1 Hour
GRADE	11	MARKS	100
	Mrs M Povall		Mrs Serafino Ms Scott Mr Gill
EXAMINER		MODERATOR	

QUESTION NUMBER	DESCRIPTION	MAXIMUM MARK	ACTUAL MARK
1	Venn Diagrams	6	
2	Probability Theory	4	
3	Combinations and Permutations	4	
4	Combinations and Permutations	11	
5	Combinations and Permutations	12	
6	Probability	8	
7	Probability	8	
8	Probability	13	
9	Normal distribution	22	
10	Normal distribution	12	
	Total Section A	100	

READ THE FOLLOWING INSTRUCTIONS CAREFULLY:

- Read all questions carefully, work neatly and show all necessary calculations.
- Do not spend too long on any single question – watch the time carefully.
- Answer all questions in the space provided.

QUESTION 1

6 MARKS

A first year university student is studying both Accounting and Economics. The probability that she will pass each subject is 0,4 and 0,55 respectively. The probability that she will pass both is 0,3.



a) Draw a Venn diagram to represent this information.

(4)

b) If she passes only one of these subjects, what is the probability that it was Accounting? (2)

QUESTION 2

4 MARKS

a) Describe, the relationship between the events A and B if $P(A | B) = P(A)$ (2)

b) Draw a Venn diagram depicting $P(A | B) = 0$ (2)



QUESTION 3

4 MARKS

In a club of 20 members, in how many ways can I choose:

a) a committee of 4 (2)



b) a president, vice-president, secretary and treasurer (2)

QUESTION 4

11 MARKS

Consider the word **HOLIDAY**.

(Letters may not be repeated in this question!)



a) How many 'words' (arrangements) can be made using these letters? (2)

b) How many 'words' (arrangements) can be made using these letters if the 'words' must start with the H and end with the Y ? (3)

c) How many 'words' (arrangements) can be made using these letters if the O and the L must be placed next to each other ? (3)

d) How many 4 letter 'words' (arrangements) can be made that end in a Y? (3)

QUESTION 5

MARKS 12

Using the digits 2,2,2,3,3,4,5:

a) How many different 7 digit numbers can be found? (4)



b) How many numbers are greater than 3 400 000? (4)



c) How many numbers are greater than 3 400 000 and divisible by 5. (4)

QUESTION 6**MARKS 8**

In a multiple choice question, five possible answers are given. Only one of the options are correct. The probability that a student will know the correct answer is 0,65. Assume that, if a student does not know the correct answer, he will guess at random.

a) What is the total probability that the correct option will be chosen? (4)



b) Given that Steven selected the correct answer, what is the probability that he guessed? (4)

QUESTION 7**MARKS 8**

It is know that 40% of students taking a Bachelor of Arts degree are male. If six students are taken at random from those taking a Bachelor of Arts degree, calculate the probability that more than four of the students are **female**.



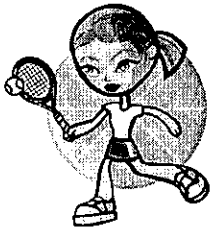
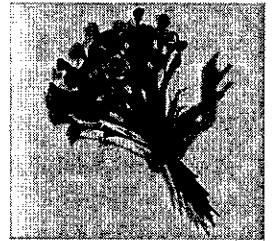
(8)



QUESTION 8

MARKS 13

- a) Consider a bunch of 5 red, 10 pink and 15 white roses. If exactly 3 flowers are taken from the bunch without replacement, what is the probability that each flower is of a different colour ? (5)



- b) A team of four tennis players are chosen is chosen from 5 girls and 6 boys. What is the probability that the team chosen has more boys than girls? (8)





QUESTION 9

MARKS 22

A rugby academy in Stellenbosch chose a squad of potential provincial rugby players. They were all weighed and it was found that their masses were normally distributed with a mean mass of 92 kg. The standard deviation was found to be 12 kg.

a) What is the probability that a player chosen at random, weighed less than 100kg? (8)

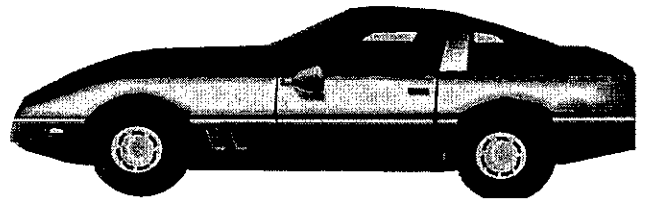


b) If 2% of the players were considered too heavy and 6% too light, what is the range of masses considered to be acceptable. (14)



QUESTION 10**MARKS 12**

The petrol consumption of a "Trambuster" (a new make of car) is known to be normally distributed about a mean of μ litres/100km with a standard deviation of σ litres/100km. If it is known that there is a probability of 0,67 that a randomly selected Trambuster has a consumption of more than 10 litres/100km and a probability of 0,86 that the consumption is less than 15,12 litres/100km. Calculate the values of μ and σ .





STATISTICS DATA SHEET

$$* P(A) = \frac{n(A)}{n(S)}$$

$$* P(B|A) = \frac{P(B \cap A)}{P(A)}$$

$$* P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$* {}^n P_r = \frac{n!}{(n-r)!}$$

$$* {}^n C_r = \binom{n}{r} = \frac{n!}{(n-r)!r!}$$

$$* P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}$$

$$* P(R=r) = \frac{\binom{p}{r} \binom{N-p}{n-r}}{\binom{N}{n}}$$

$$* z = \frac{X - \mu}{\sigma}$$

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$z = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y}}}$$

$$b = \frac{n \sum(xy) - \sum x \sum y}{n(\sum x^2) - (\sum x)^2}$$

$$b = \frac{\sum xy - n\bar{x}\bar{y}}{\sum x^2 - n(\bar{x})^2}$$

$$b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

* RELEVANT TO GRADE 11

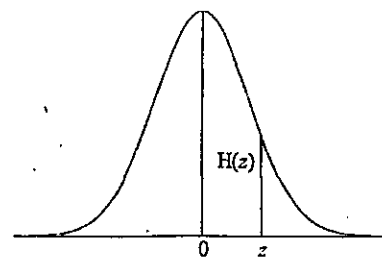
NORMAL DISTRIBUTION TABLE

Areas under the Normal Curve

$$H(z) = \frac{1}{\sqrt{2\pi}} \int_0^z e^{-\frac{1}{2}x^2} dx$$

$$H(-z) = H(z), H(\infty) = \frac{1}{2}$$

Entries in the table are values of $H(z)$ for $z \geq 0$.



z	,00	,01	,02	,03	,04	,05	,06	,07	,08	,09
0,0	,0000	,0040	,0080	,0120	,0160	,0199	,0239	,0279	,0319	,0359
0,1	,0398	,0438	,0478	,0517	,0557	,0596	,0636	,0675	,0714	,0753
0,2	,0793	,0832	,0871	,0910	,0948	,0987	,1026	,1064	,1103	,1141
0,3	,1179	,1217	,1255	,1293	,1331	,1368	,1406	,1443	,1480	,1517
0,4	,1554	,1591	,1628	,1664	,1700	,1736	,1772	,1808	,1844	,1879
0,5	,1915	,1950	,1985	,2019	,2054	,2088	,2123	,2157	,2190	,2224
0,6	,2257	,2291	,2324	,2357	,2389	,2422	,2454	,2486	,2517	,2549
0,7	,2580	,2611	,2642	,2673	,2704	,2734	,2764	,2794	,2823	,2852
0,8	,2881	,2910	,2939	,2967	,2995	,3023	,3051	,3078	,3106	,3133
0,9	,3159	,3186	,3212	,3238	,3264	,3289	,3315	,3340	,3365	,3389
1,0	,3413	,3438	,3461	,3485	,3508	,3531	,3554	,3577	,3599	,3621
1,1	,3643	,3665	,3686	,3708	,3729	,3749	,3770	,3790	,3810	,3830
1,2	,3849	,3869	,3888	,3907	,3925	,3944	,3962	,3980	,3997	,4015
1,3	,4032	,4049	,4066	,4082	,4099	,4115	,4131	,4147	,4162	,4177
1,4	,4192	,4207	,4222	,4236	,4251	,4265	,4279	,4292	,4306	,4319
1,5	,4332	,4345	,4357	,4370	,4382	,4394	,4406	,4418	,4429	,4441
1,6	,4452	,4463	,4474	,4484	,4495	,4505	,4515	,4525	,4535	,4545
1,7	,4554	,4564	,4573	,4582	,4591	,4599	,4608	,4616	,4625	,4633
1,8	,4641	,4649	,4656	,4664	,4671	,4678	,4686	,4693	,4699	,4706
1,9	,4713	,4719	,4726	,4732	,4738	,4744	,4750	,4756	,4761	,4767
2,0	,4772	,4778	,4783	,4788	,4793	,4798	,4803	,4808	,4812	,4817
2,1	,4821	,4826	,4830	,4834	,4838	,4842	,4846	,4850	,4854	,4857
2,2	,4861	,4864	,4868	,4871	,4875	,4878	,4881	,4884	,4887	,4890
2,3	,48928	,48956	,48983	,49010	,49036	,49061	,49086	,49111	,49134	,49158
2,4	,49180	,49202	,49224	,49245	,49266	,49286	,49305	,49324	,49343	,49361
2,5	,49379	,49396	,49413	,49430	,49446	,49461	,49477	,49492	,49506	,49520
2,6	,49534	,49547	,49560	,49573	,49585	,49598	,49609	,49621	,49632	,49643
2,7	,49653	,49664	,49674	,49683	,49693	,49702	,49711	,49720	,49728	,49736
2,8	,49744	,49752	,49760	,49767	,49774	,49781	,49788	,49795	,49801	,49807
2,9	,49813	,49819	,49825	,49831	,49836	,49841	,49846	,49851	,49856	,49861
3,0	,49865	,49869	,49874	,49878	,49882	,49886	,49889	,49893	,49896	,49900
3,1	,49903	,49906	,49910	,49913	,49916	,49918	,49921	,49924	,49926	,49929
3,2	,49931	,49934	,49936	,49938	,49940	,49942	,49944	,49946	,49948	,49950
3,3	,49952	,49953	,49955	,49957	,49958	,49960	,49961	,49962	,49964	,49965
3,4	,49966	,49968	,49969	,49970	,49971	,49972	,49973	,49974	,49975	,49976
3,5	,49977									
3,6	,49984									
3,7	,49989									
3,8	,49993									
3,9	,49995									
4,0	,49997									